AI Safety HW 1 Report

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Experiment Goal: We tested whether Qwen2.5-3B-Instruct could solve GSM8K math problems in Mario's voice without losing accuracy. In our September 2026 SFT evaluation, fine-tuning made answers more similar to Mario references, but test accuracy fell from 73.0% to 66.6%. Plain math fine-tuning also reduced accuracy, so we could not attribute the decline specifically to the persona.

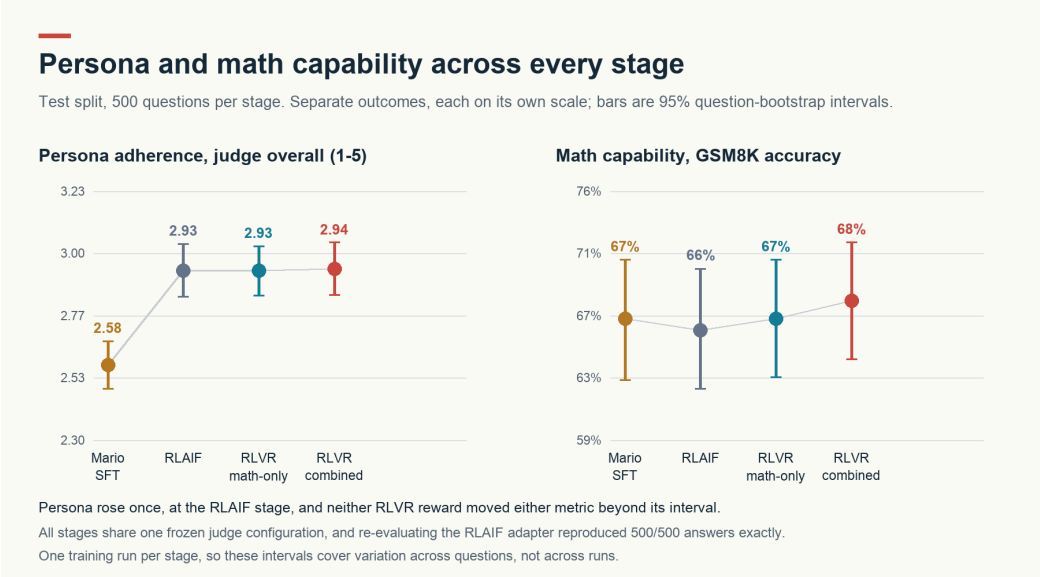
Hypothesis 1: A persona learned through fine-tuning is stronger than one produced by prompting. We tested this by comparing Mario SFT with the original model given a Mario instruction.

Hypothesis 2: Persona training lowers math accuracy. We compared Mario SFT with both the original model and a model fine-tuned on plain GSM8K answers. The plain-SFT control helps distinguish a possible cost of persona training from a broader effect of fine-tuning.

Method: The project has three stages: supervised fine-tuning (SFT), reinforcement learning from AI feedback (RLAIF), and reinforcement learning with verifiable rewards (RLVR). SFT teaches the model from example answers, RLAIF uses AI judgments of persona quality, and RLVR uses a rule-based math checker. We evaluate voice and accuracy separately because a model can improve at one while getting worse at the other.

The bulk of the reflection and summary is in the following section. For additional context and reflection, you can read the sections afterwards that go into more depth as to what the set up, training, evaluation, and results looked like for each phase of our project.

Cross-Phase Comparison & Reflection



Overall, the RLAIF and RLVR-trained models scored higher on persona adherence than the Mario SFT model, while math capability showed no significant difference among those three. As per the above image, these two metrics appear independent, and essentially all of the persona gain came from a single stage: RLAIF raised the judge's score from 2.58 to 2.93 out of 5 on the test set, and our RLVR runs didn't seem to improve the model in any significant manner.

Our second hypothesis was that persona training would come at the cost of mathematical accuracy. The intuition was centered on model capacity: a fixed-capacity model that gets better at sounding like Mario might have less capability to also be good at math, so gains in one objective would show up as losses in the other. The data doesn't necessarily support this hypothesis either. Accuracy fell once from 73.0% for the base model to 66.6% after Mario SFT. However, plain-GSM8K SFT, trained with no persona at all, fell further still to 65.2%, so the loss came from supervised fine-tuning in general rather than from learning a character. Perhaps our second hypothesis might've been correct for a smaller model, but the two capabilities didn't seem to tradeoff here for the observed model size. These results may also be specific to Mario, whose persona is identifiable in the use of particular phrases rather than how they talk about subjects like math or solve math problems.

While RLAIF substantially improved the persona rating of our Mario math answerer, there were opportunities to improve its training. For context, of the 1,000 training questions, only 773 of their responses were selected as pairs for training via RLAIF as the 227 discarded questions didn't satisfy our criteria, which primarily consisted of two things: (1) both answers must be mathematically correct, isolating the difference in persona, and (2) the gap in persona scores must be at least 1. More training pairs might've been retained had we considered increasing the number of sampled responses from 6 to 10 or introducing more distinction in the LLM judge's evaluations via an improved system prompt. At the same time, RLAIF could have had better training if we were more selective in the chosen pairs by increasing the minimum gap in persona scores from 1 to 2. That way, the signal is much clearer as to whether a LLM's generated response is favorable or not. Some additional tests could've been performed – such as training with only 500 pairs or changing the minimum persona gap – to better isolate the drivers in our RLAIF's performance.

Based on our tests, we didn't find evidence of the AI judge getting gamed. Our initial observation of various catchphrases such as "Wahoo" or "It's-a-me" initially led us to restrict these types of expressions in our persona's *constitution* and track the number of catchphrase-only answers. An agent that games the judge could've essentially spammed these catchphrases without actually sounding like Mario when answering the math question. However, the rate of catchphrase-only responses dropped from 54.8% to 42.4% during the RLAIF stage and remained there during RLVR training. Another avenue that could have been leveraged for an agent to game the judge is to provide longer responses. However, we tracked the length of responses and found that for answer pairs within 10% of each other in word count, the persona gain by RLAIF was 0.179 relative to Mario SFT, indicating that gaming via response length didn't

seem to take place. For these reasons, we find that our set up did a good job in tracking and mitigating potential avenues for AI-judge gaming.

Overall, improvements in the persona score didn't seem to systematically trade off with any particular metric that we tracked. However, a test that we could incorporate moving forward is to take a random sample of responses and judge-assigned scores and have a human review it. Since the same LLM-based judge family that scored the model's answers also generated the RLAIF training preferences and the RLVR persona reward, its ratings partly reflect how well the model learned to satisfy its own grader rather than how in-character it sounds to an independent reader. Having a human grade a sample of responses or a model's responses to a held out test-set would help ensure that, in the end, the training is geared towards satisfying an independent evaluation. The 50 pairs (located in **brown-ogbogu-tantia > reports > persona_review > review.md**) illustrate how this might look in practice.

Upon reflection of our RLVR run, our learning rate and KL penalty might've been too conservative to produce behavior distinct from RLAIF: 32 updates at a $1e^{-6}$ learning rate and KL penalty of 0.02 might've been too weak to see meaningful changes in our model over the course of training. We also suspected that the combined reward function hindered the model's ability to improve persona. Persona can add at most 0.25 to a correct answer's score, while correctness itself is worth 1.0, so the persona term only has room to act inside the 215 groups where all four sampled answers were already correct. We built a sweep over weights 0.25, 0.5, 0.75, and 1 for the persona weighting to test this, but decided not to run it due to time and cost. At the same time, the math-only RLVR tests performed similarly to RLAIF and the combined-RLVR training despite not having the persona term, which might indicate an issue in the number and pace of updates rather than the weighting of the persona term. Raising the update count or learning rate and lowering the KL penalty term might've helped in training the combined-RLVR reward.

Phase 1: SFT

SFT taught the model to use more Mario-like language while solving math problems, but it did not achieve our accuracy goal. We evaluated four conditions on GSM8K on September 12, 2026: the original model, the original with a Mario prompt, plain math SFT, and Mario SFT.

Setup:

We used paired versions of 3,000 GSM8K training examples so the two SFT models learned from the same questions. One version kept the original explanations; the other rewrote every explanation in Mario's voice. We also set aside 500 validation questions and 500 questions from the official test split.

Rewriting the math answers directly gave the model examples of the behavior we wanted at evaluation: explaining a calculation in character. A mixture of Mario dialogue and plain math solutions would leave the model to combine those behaviors itself, while some math examples

would still teach a neutral voice. The paired design also made the plain-SFT comparison easier to interpret. We did not test the mixed-data alternative, so we cannot claim that our choice was better.

Our first approach added stock Mario phrases to otherwise unchanged solutions. We rejected it because the voice had little connection to the problem. Astra and Sonnet then rewrote the explanations so Mario reacted to the situation and narrated the calculation. For example, a restaurant problem began, "A family dinner, now that's-a worth sitting down for!" The rewrites kept the question and final numeric answer. We checked numerical preservation automatically and reviewed samples for meaning and voice.

The validation and test questions stayed out of model training. Validation could have guided settings or checkpoint selection, but we made no changes after seeing its results. In this experiment, it therefore served as an additional evaluation set. One limitation is that the Astra rewrite workers saw three validation examples as style guidance, which weakens the separation for the persona comparison.

Training:

We trained two LoRA adapters on Qwen2.5-3B-Instruct, one for plain answers and one for Mario rewrites. LoRA updates a small set of added parameters rather than all of the model's weights, making the comparison feasible with limited compute.

The training code calculates loss only on the assistant's response. The question supplies context, while the learning signal comes from predicting the explanation and final answer. This trains Mario SFT on the style throughout a worked solution, including how it describes each calculation. Both adapters use LoRA rank 16, alpha 32, and dropout 0.05.

We supplied no Mario instruction during SFT and evaluated both adapters without one. This tests whether the adapter learned the voice. The prompted-original condition provided the alternative: ask the unmodified model to speak as Mario without training it.

Evaluation:

Mario SFT improved reference similarity but reduced math accuracy relative to the original model. We generated answers for all four conditions on the same questions, using greedy decoding and a shared instruction to explain the solution and provide a machine-readable final answer. We ran the evaluation on a cloud GPU.

Math accuracy. Both SFT models performed worse than the original on validation and test. On test, Mario SFT reached 66.6%, compared with 73.0% for the original and 65.2% for plain SFT. On validation, the corresponding scores were 73.2%, 85.2%, and 74.4%.

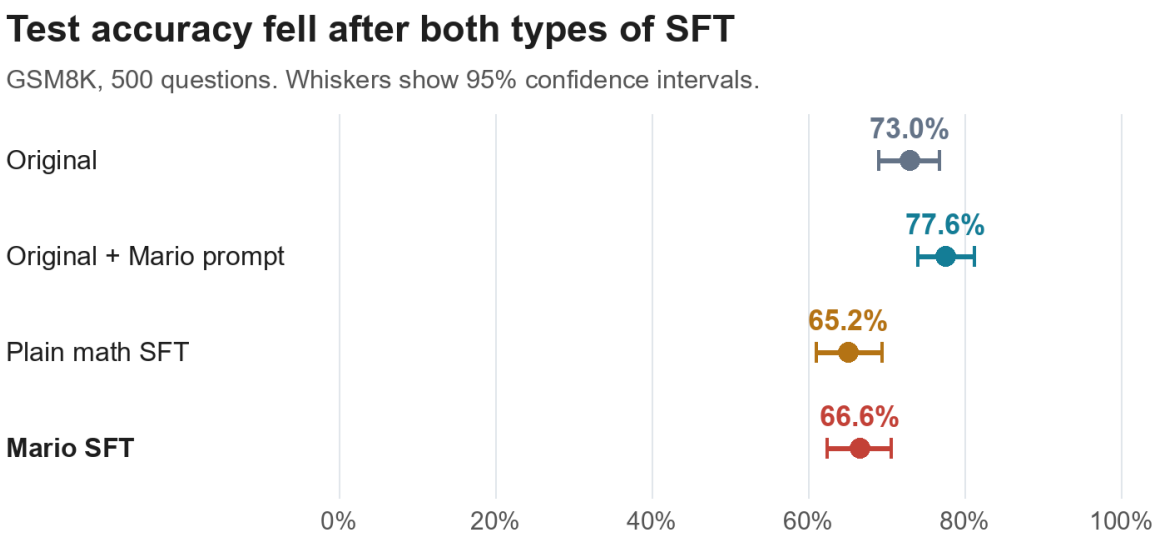


Figure 1. Test accuracy across the four conditions. Each model answered the same 500 questions.

Mario SFT lost 6.4 percentage points against the original on the test, with a 95% confidence interval of -11.0 to -2.0. Against plain SFT, however, Mario SFT gained 1.4 points, with an interval of -3.2 to +6.0. That comparison was inconclusive. Hypothesis 2 held relative to the original model, but we found no clear extra penalty from persona training.

Prompting had the highest test accuracy, at 77.6%, despite requiring no weight updates. It did not lead on validation, so we should not assume that a Mario prompt generally improves math. Another unexpected result was that Mario SFT made final answers easier to parse: format success rose from 85.8% to 99.4% on test, even as accuracy fell. The model learned the output format more reliably without solving more problems correctly.

Persona similarity. Mario SFT produced answers closer to the Mario references than prompting alone. On the 500 validation questions, mean cosine similarity was 0.4021 for Mario SFT and 0.3593 for the prompted original. Mario SFT's advantage over prompting was 0.0428, with a 95% confidence interval of 0.0359 to 0.0499.

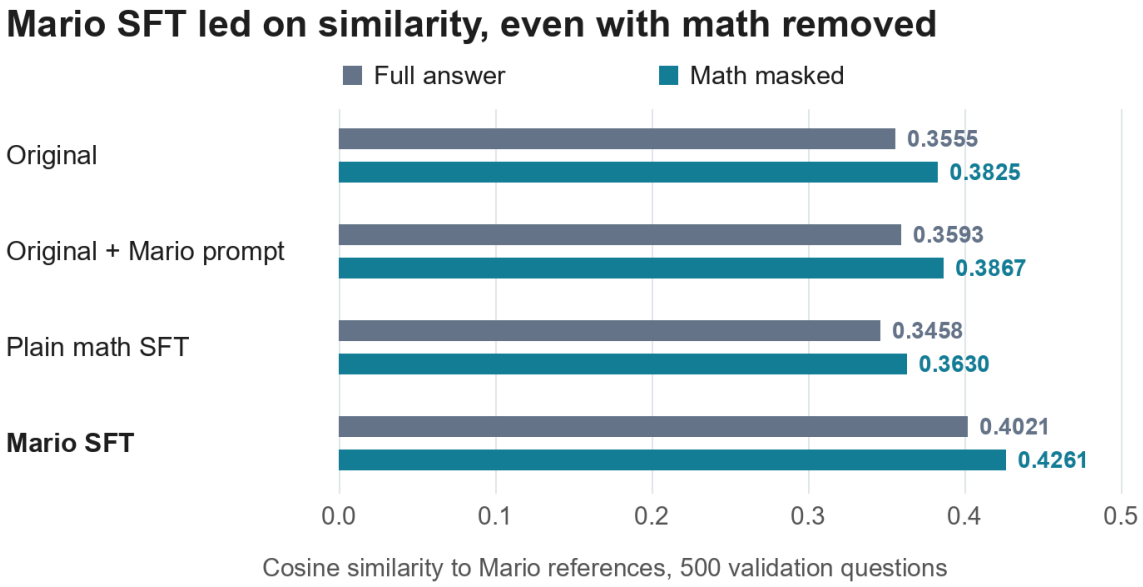


Figure 2. Similarity to Mario reference answers, before and after removing math notation.

We used character n-gram TF-IDF to measure shared wording. To check whether repeated calculations explained the gain, we removed numbers, operators, calculator annotations, and final-answer lines before repeating the comparison. With math masked, Mario SFT still led prompting, 0.4261 to 0.3867. This supports Hypothesis 1 for our prompt and similarity measure, though shared wording is only a proxy for convincing character voice. We did not collect blinded persona ratings for this SFT comparison.

The outputs showed why voice and correctness need separate checks. On one window-panel problem, Mario SFT stayed in character but answered 20 instead of 80 after omitting the final multiplication. On a cake problem, it answered correctly and narrated the calculation in character, but added a detail about frosting that the question never mentioned. These examples illustrate the failure modes; they do not measure how often they occurred.

SFT produced a stronger textual resemblance to Mario, but we did not meet the capability-improvement goal. The plain-SFT decline suggests a broader fine-tuning problem worth investigating, rather than a demonstrated conflict between Mario's voice and math. We scored final answers rather than every reasoning step, and counted unparseable answers as wrong. This single-run experiment cannot establish the cause of the decline.

Phase 2: RLAIF

After running SFT, reinforcement learning from artificial intelligence feedback (RLAIF) was used to additionally fine-tune the model for improved persona and math capability. RLAIF strengthened the Mario persona as rated by an AI judge, with no clear change in math accuracy. On 500 test questions, the overall persona score rose from 2.58 to 2.93 out of 5.

Setup:

Direct Preference Optimization (DPO) was used to train a new version of the model. DPO was chosen due to its efficient use of compute and API spend.

The initial dataset was generated by having the Mario SFT model answer 1,000 training questions six times each, with a temperature of 1. This yielded 6,000 total answers. These answers were then subject to a math check to grade each as correct or incorrect, and then an AI judge (Claude Haiku 4.5) scored the persona of each answer from 1 to 5. The Persona for each answer was rated on voice consistency, naturalness, restraint, and overall performance. It also flagged responses that were comprised entirely of stock catchphrases.

Criterion	**1**	**3**	**5**
Voice: sounds like Mario	Neutral assistant, no Mario	Some Mario touches, but could be any cheerful character	Unmistakably Mario
Consistency: voice lasts the whole answer	Mario in one line only	Mario in most of the answer, with neutral stretches	Mario from the first line to the final answer
Naturalness: reads like a real character	Mechanical filler between equations	Mostly readable, some lines feel bolted on	Flows as one voice and reacts to the actual problem
Restraint: character helps, doesn't overwhelm	Catchphrase spam, or no persona at all	Heavy-handed, but the math stays clear	Light touch, math easy to follow
Overall: one judgment, not an average	"Not at all"	"Sort of"	"Yes, that's him"

Figure 3: Persona Judge rubric criterion

After checking and scoring the 6,000 responses, we were left with 773 pairs. Any questions that were answered incorrectly or had less than a 1-point difference in persona score were removed from the dataset. This was done in an effort to protect the math performance of the model.

Training:

We train the model with DPO to raise the probability of the model outputting a preferred answer to a less preferred one. This training continued with the Mario SFT LoRA adapter. We used 1 epoch of 49 steps, an effective batch size of 16, and a seed of 42. The training occurred on an A100 GPU through Runpod.

Evaluation:

Once the train was complete we reran the Mario SFT alongside RLAF in the same environment to evaluate the persona and math accuracy

Persona. RLAIF raised the overall persona score on both test and validation. On test it rose from 2.58 to 2.93, with a 95% confidence interval for the gain of 0.24 to 0.46. On validation it rose from 2.63 to 3.12, with an interval of 0.38 to 0.60. On test, RLAIF scored higher on 154

questions, tied on 278 and scored lower on 68. Each of the four rubric scores rose by between 0.26 and 0.49 points. The share of catchphrase-only answers fell from 54.8% to 42.2% on test and from 55.8% to 37.0% on validation, so the gain did not come from adding stock phrases.

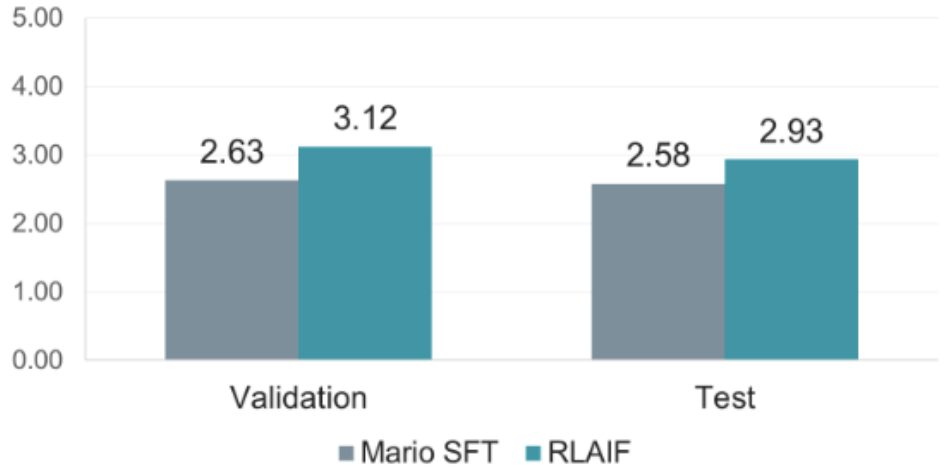


Figure 4: RLAIF Persona Evaluation Score

RLAIF also wrote longer answers, averaging 105 words against 87 on the test. This may be one factor that is contributing to the improved persona scores.

Math capability. DPO did not meaningfully impact the math accuracy. Test accuracy moved from 67.0% to 66.2%, while the validation moved from 75.2% to 76.6%. We cannot conclude that accuracy was preserved, but that we did not detect additional cost in accuracy due to persona training via DPO.

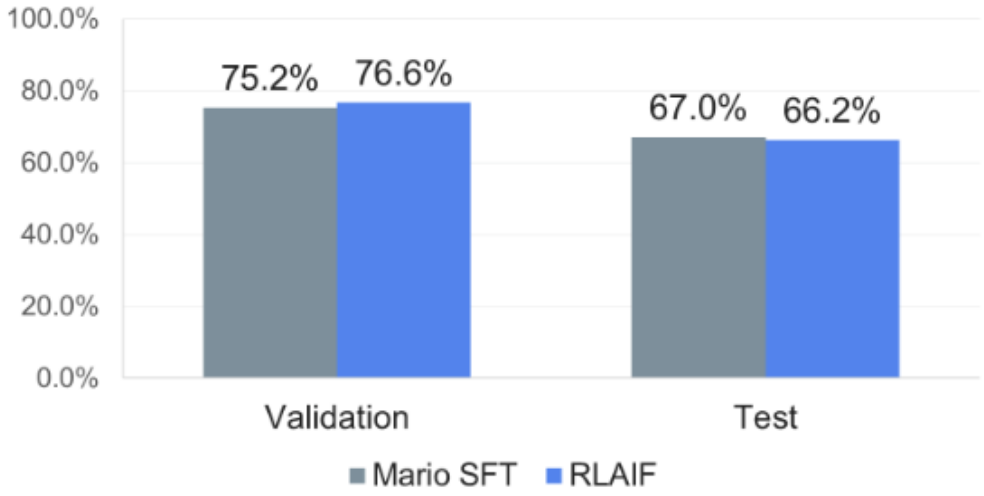


Figure 5: RLAIF Math Capability Score

Phase 3: RLVR

Setup: In our last phase, we wanted to test two hypotheses:

Hypothesis 1 (Math-Only Reward): Reinforcement Learning with Verifiable Rewards (RLVR) would improve the performance of our Qwen-based LLM agent for math accuracy but cause the model to perform worse for persona, relative to our RLAIF baseline. Intuitively, the math-only RLVR reward function provides clear signal for math accuracy improvement, but fails to signal improvements in persona adherence.

- $Reward = Correct$

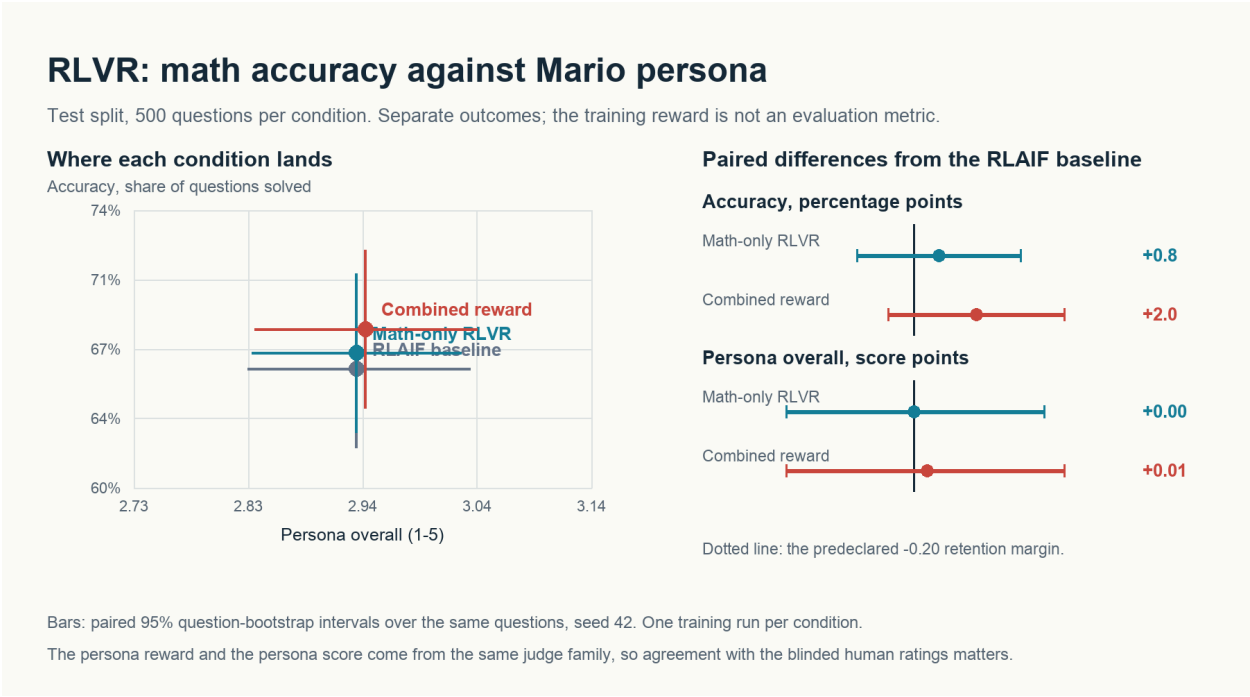
Hypothesis 2 (Combined Reward): The combined reward for RLVR would perform better than the math-only RLVR reward and RLAIF baseline across both math accuracy and overall persona score. The intuition for this is that the combined reward function captures both math accuracy and persona adherence explicitly, which both the math-only RLVR reward function and RLAIF fail to do. By clearly specifying this reward function, the agent gets a very clear signal for how to improve across its training runs.

- $Reward = Correct * (1 + 0.25 * Persona)$

Training: The RLAIF baseline and the GRPO performance under both RLVR and RLAIF/RLVR rewards were trained using 512 randomly chosen training questions. In sum, there were 32 updates and 2,048 sampled answers per run. Training took place on Runpod with a A100-80GB. The learning rate was set to $1e^{-6}$ and the KL penalty was 0.02.

Evaluation: We stayed consistent in the evaluation approach as the previous trials, so Claude Haiku 4.5 had used the same rubric as during training when scoring all answers provided by the three adapters (RLAIF baseline and GRPO with both RLVR rewards).

Results



In general, we found that there wasn't enough evidence to conclude that the model adapters trained with RLVR-based rewards performed differently from the RLAIF baseline. The above image displays these findings on the right side, where the confidence interval for the Math-only RLVR and Combined reward includes the RLAIF's performance (across math accuracy and overall persona scores). For a detailed numerical accounting, please see the document RLVR_RESULTS.md in the project's reports folder.

Given our confidence intervals, it seems that we don't have enough data to prove or disprove our hypotheses. With that said, Math-only RLVR improved math accuracy by +0.8 [-1.8, +3.4] and the combined reward improved the math accuracy by +2.0 [-0.8, +4.8]. Persona moved +0.000 [−0.074, +0.076] under the math-only reward and +0.008 [−0.074, +0.088] under the combined reward.